

Progress Test 3.B — Thermo-S1

3.B · The ideal gas and Clapeyron's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Units

(1 mark)

Express the unit of the gas constant R , $\text{J K}^{-1} \text{mol}^{-1}$, in SI base units.

$1 \text{ J} = 1 \text{ N m} = 1 \text{ kg m}^2 \text{ s}^{-2}$	$[R] = \text{kg m}^2 \text{ s}^{-2} \text{ K}^{-1} \text{mol}^{-1}$
---	---

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $\text{kg m}^2 \text{ s}^{-2} \text{ K}^{-1} \text{mol}^{-1}$.

Exercise 2. The ideal gas — definition and Clapeyron's law

(5 marks)

Define the ideal gas model. Write Clapeyron's law, giving the SI unit of every quantity and the value of R .

The ideal gas is the limit model towards which <i>all</i> gases tend when their particle density n/V is low enough. Its particles have no interactions other than collisions with the walls of the container. Clapeyron (1834): $pV = nRT$ p in Pa, V in m^3, n in mol, T in K, $R = 8.314 \text{ J K}^{-1} \text{mol}^{-1}$.

Marking

①
0 1 2
☐ ☐ ☐

②
0 1 2
☐ ☐ ☐

③
0 1
☐ ☐

Marking scheme

- ① 2 pts Low-density limit (1 pt); no interactions between particles (1 pt).
 ② 2 pts $pV = nRT$.
 ③ 1 pt The five units, with T in kelvins, and $R \approx 8.31 \text{ SI}$.

Exercise 3. Controlled and free variables

(3 marks)

For a hot-air balloon (opening at the bottom, rigid envelope), the variables held fixed are:

- ☐ n and V ☒ p and V ☐ n and T ☐ n and p

Why: Tab. 5: $p = p_{\text{atm}}$ (mechanical equilibrium) and V is fixed by the rigid envelope. n and T are free: heating drives air out.

A weather balloon (sealed, flexible) rises; the outside pressure decreases while its temperature stays about the same. Its volume:

- ☒ increases ☐ decreases ☐ stays the same ☐ becomes zero

Why: n is fixed and $p = p_{\text{atm}}$ decreases. With $pV = nRT$ and T about constant, V increases.

1.0 mol of gas at 300 K and 1.0×10^6 Pa occupies 2.5 L. With $RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$, its compressibility factor $Z = pV/(nRT)$ is about:

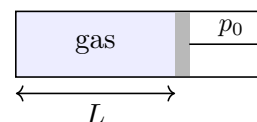
- ☐ 0.1 ☒ 1.0 ☐ 10 ☐ 2.5

Why: $Z = \frac{1.0 \times 10^6 \times 2.5 \times 10^{-3}}{2.5 \times 10^3} = 1.0$: the gas behaves as an ideal gas. (Compare the worked example of the notes: 1 mol of air in 2 L at 2 bar and 25 °C is far from ideal.)

Exercise 4. Piston and cylinder

(5 marks)

A horizontal cylinder of cross-section S contains n moles of air, treated as an ideal gas. The piston is initially at L_1 and the temperature is T_1 .



(1) Compute p_1 . (2) The gas is compressed isothermally to L_2 : compute p_2 . (3) The piston is locked at L_2 and the gas is heated to T_3 : compute p_3 . (4) The piston is released against the atmosphere p_0 , at $T_4 = T_3$: compute V_4 and L_4 .

$$S = 50 \text{ cm}^2 \quad n = 0.10 \text{ mol} \quad L_1 = 50 \text{ cm} \quad L_2 = 25 \text{ cm} \quad T_1 = 300 \text{ K}$$

$$T_3 = 450 \text{ K} \quad p_0 = 1.0 \times 10^5 \text{ Pa} \quad RT \approx 2.5 \times 10^3 \text{ J mol}^{-1} \text{ at } 300 \text{ K}$$

$$(1) V_1 = SL_1 = 50 \times 10^{-4} \times 0.50 = 2.5 \times 10^{-3} \text{ m}^3$$

$$p_1 = \frac{nRT_1}{V_1} = \frac{0.10 \times 2.5 \times 10^3}{2.5 \times 10^{-3}} \quad p_1 = 1.0 \times 10^5 \text{ Pa}$$

$$(2) n \text{ and } T \text{ fixed: } p_1 V_1 = p_2 V_2, \text{ so } p_2 = p_1 \frac{L_1}{L_2} \quad p_2 = 2.0 \times 10^5 \text{ Pa}$$

$$(3) n \text{ and } V \text{ fixed: } \frac{p}{T} = \text{const.}, \text{ so } p_3 = p_2 \frac{T_3}{T_1} = 2.0 \times 10^5 \times \frac{450}{300}$$

$$p_3 = 3.0 \times 10^5 \text{ Pa}$$

$$(4) n, T \text{ and } p = p_0 \text{ fixed: } V_4 = \frac{nRT_4}{p_0}, \text{ with}$$

$$RT_4 = 1.5 \times 2.5 \times 10^3 = 3.75 \times 10^3 \text{ J mol}^{-1}:$$

$$V_4 = \frac{0.10 \times 3.75 \times 10^3}{1.0 \times 10^5} \quad V_4 = 3.75 \times 10^{-3} \text{ m}^3$$

$$L_4 = \frac{V_4}{S} = \frac{3.75 \times 10^{-3}}{5.0 \times 10^{-3}} \quad L_4 = 0.75 \text{ m}$$

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	
⑤		
0	1	

Marking scheme

- ① 1 pt $p_1 = 1.0 \times 10^5 \text{ Pa}$ (with S and L_1 converted to SI).
- ② 1 pt $p_2 = 2.0 \times 10^5 \text{ Pa}$.
- ③ 1 pt $p_3 = 3.0 \times 10^5 \text{ Pa}$.
- ④ 1 pt $V_4 = 3.75 \times 10^{-3} \text{ m}^3$ (3.75 L).
- ⑤ 1 pt $L_4 = 75 \text{ cm}$.

Examiner's tip

As in Tab. 5 of the notes, first list which variables are fixed at each step: then the right law (pV , p/T or V/T constant) follows at once.

Your score: / 14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = / 20